Bio301

Essential of genetics

FINAL TERM OBJECTIVE + SUBJECTIVE

Lion group of Zoologist

GROUP PROPERTIES

- Separate Group For Each Subject
- Separate Group For Discussion
- vedio Lectures
- Ideal Assignments
- Ideal gdbs

اس گروپ میں ویڈیو لیکچرز، آئیڈیل اسائنمنٹس
اور آئیڈیل GDB مہیا کی جاتی ہے

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1st Excellent Zoologist Group for 1st Semester
And Lion Group for 2nd Semester

BIO-301
PAST PAPERS

A point mutation

A **point mutation** or substitution is a genetic **mutation** where a single nucleotide base is changed, inserted or deleted from a sequence of DNA or RNA. **Point mutations** have a variety of effects on the downstream protein product—consequences that are moderately predictable based upon the specifics of the **mutation**.

Complications Of Inheritance?

Complications to inheritance patterns: Multiple genes - epistasis, polygenic traits Genes & the environment - sex-influenced traits, incomplete penetrance.

Complications : A common recessive inheritnace can give dominant pattern Autosomal dominant inheritance with non-penetrance, Autosomal dominant inheritance with variable expression.

Mutation Effects In Functioning Of Genes?

Some mutations don't have any noticeable effect on the phenotype of an organism. This can happen in many situations: perhaps the mutation occurs in a stretch of DNA with no function, or perhaps the mutation occurs in a **protein**-coding region, but ends up not affecting the amino acid sequence of the **protein**.

Causes Of Epigenetics?

Stress Hormone **Causes Epigenetic** Changes. Researchers found that chronic exposure to a stress hormone **causes** modifications to DNA in the brains of mice, prompting changes in gene expression

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Function Of Pedigree?

A pedigree may be used to establish the probability of a child having a particular disorder or condition. It may be used to discover where the genes in question are located (x, y, or autosome chromosome), and to determine whether a trait is **dominant** or recessive.

Define Epigenetics?

Epigenetics is the study of heritable phenotype changes that do not involve alterations in the DNA sequence

RAPD And ALFPDifference?

RAPD is Random Amplification of Polymorphic DNA'. It is a type of PCR, but the segments of DNA that are amplified are random.

AFLP is a PCR-based tool used **in genetics** research, DNA fingerprinting, and **in**

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the practice of **genetic** engineering. Developed **in** the early 1990s by Keygene, **AFLP** uses restriction enzymes to digest genomic DNA, followed by ligation of adaptors to the sticky ends of the restriction fragments.

Forward, Reverse, And Neutral Mutations: 3marks

Forward mutation: Mutation that alters the wild type phenotype into mutant phenotype.
Reverse mutation: Mutation that changes a mutant phenotype back into the wild type.
Neutral mutation: Mutation that alters the amino acid sequence of the protein but does not change its functionasreplacedaminoacidischemicallysimilarortheaffected AAhaslittle influenceonproteinfunction

Point Mutations And Enlist 3 Structural Changes In Chromosomes?

A point mutation or substitution is a genetic mutation where a single nucleotide base is changed, inserted or deleted from a sequence of DNA or RNA. Point mutations have a variety of effects on the downstream protein product—consequences that are moderately predictable based upon the specifics of the mutation. Deletion, duplication, translocation, transcription.

Structural Change in the Structure of Chromosomes.....Chromosomes are the vehicle of hereditary material or genes. Any alteration, addition or deletion of chromosomal part leads toalteration of number, position or sequence of genes in thechromosome.

Complications In Inheritance?

Epistasis, polygenic traits Genes & the environment - sex-influenced traits, incomplete penetrance.

Modification Of Epigenetica

Epigeneticmodification- noncoding RNAisanewmechanismforgeneregulation. RNA which is not used for making proteins can be cleaved and used to inhibit protein-coding RNAs (siRNAs, microRNAs). Three common types of modification - Cytosine methylation Histonesmodifications siRNAs

Mutation Alter Genes Expression?

Non-coding mutations: Promoter/enhancer element trinucleotide repeats in 5 and 3'UTRs Splice sites. Effect of mutation on gene function. Null allele (loss of function) - no gene product.
Hypomorph– decreasedactivity. Gainoffunction- increasedactivity. Dominantnegative-

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antagonizesnormal product. Neomorph - novel activity. Predicting that a gene product won't do the job. Deletion, nonsense, frameshift of sequence is deleterious. Mutation in splice site usually has sever effects. Missense mutations depend on location in protein. Is it non-conservative. Is AA conserved in evolution.

RFLP And RAPD?

RFLP is a technique that exploits variations in homologous DNA sequences. **RFLP** analysis was an important tool in genome mapping, localization of genes for genetic disorders, determination of risk for disease, and paternity testing.

RAPD is Random Amplification of Polymorphic DNA'. It is a type of PCR, but the segments of DNA that are amplified are random.

Genetic Mapping?

Gene mapping describes the methods used to identify the locus of a gene and the distances between genes. The essence of all genome mapping is to place a collection of molecular markers onto their respective positions on the genome.

Sickle cell anemia?

Sickle cell anemia (**sickle cell disease**) is a disorder of the blood caused by an inherited abnormal hemoglobin (the oxygen-carrying protein within the red blood **cells**). The abnormal hemoglobin causes distorted (sickled) red blood **cells**. The sickled red blood **cells** are fragile and prone to rupture.

Point mutation and cause?

A **point mutation** or substitution is a genetic **mutation** where a single nucleotide base is changed, inserted or deleted from a sequence of DNA or RNA. **Point mutations** have a variety of effects on the downstream protein product—consequences that are moderately predictable based upon the specifics of the **mutation**.

Cause:

The protein may lose its function, which can **result in** a disease in the organism. For example, sickle-cell disease is **caused** by a single **point mutation** (a missense **mutation**) in the beta-hemoglobin gene that converts a GAG codon into GUG, which encodes the amino acid valine rather than glutamic acid.

Mutation nomenclature?

Standard **nomenclature** based on genomic DNA reference sequences and RNA reference sequences is not shown. "Genomic DNA reference sequence" simply indicates any human DNA sequence in the database that is not based on a cDNA sequence. Example of standard

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mutation nomenclature based on a coding DNA sequence.

VNTR and application?

VNTR are short nucleotide sequences organized as a tandem repeats on genomes.Found on many chromosomes, Show variations in length between individuals. **Applications of VNTR** Microbiology DNA fingerprinting Genetic diversity Forensics Mapping of genomes Breeding patterns of wild or domesticated animals .

Single And Multifactorial Mutation?

Single gene disorders are caused by DNA changes in one particular **gene**, and often have predictable inheritance patterns. Over 10,000 human disorders are caused by a change, known as a **mutation**[?], in a **single gene**[?]. These are known as **single gene** disorders

Multifactorial Disorders. Researchers are learning that nearly all conditions and **diseases** have agenetic component. Some disorders, such as sickle cell disease and cystic fibrosis, are caused by **mutations** in a single gene. Conditionscaused by manycontributingfactorsare calledcomplexor **multifactorial** disorders.

5 Complication Inheritance Pattern?

Complications to inheritance patterns Multiple genes - epistasis, polygenic traits Genes & the environment - sex-influenced traits, incomplete penetrance

Autosomal Domenet Disordered?

A single abnormal gene on one of the first 22 nonsex (**autosomal**) chromosomes from either parent can causean **autosomal** disorder.**Dominant** inheritance means an abnormal genefromoneparent can cause disease .....Examples of **autosomal dominant disorders** include Marfan syndrome and neurofibromatosis type 1.

Function Of STR?

Short Tandem Repeat (**STR**) analysis is one of the most useful methods in molecular **biology** which is used to compare specific loci on DNA from two or more samples. A short tandem repeat is a microsatellite, consisting of aunitoftwo to thirteennucleotides repeatedhundredsoftimes in arowonthe DNA strand.

During DNA Extraction How Proteins Are Denatured?

Adding proteinase

Define Locus?

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A **locus** (plural **loci**) in **genetics** isafixedpositiononachromosome, likethepositionofa geneoramarker (**genetic** marker). Each chromosome carries many genes; human's estimated 'haploid' protein coding genes are 19,000–20,000, on the 23 different chromosomes.

Sex Linked Traits And Disorder?

Genes on the X or Y chromosome are called **sex-linked .....**For example, an X-**linked** recessive allele in humans causes haemophilia, which is much more common in males than females because males are hemizygous and therefore express the**trait** when they inherit one mutant allele

Define Multiple Alleles With Example?

Multiple alleles Three or more alternative forms of a gene (**alleles**) that can occupy the same locus. However,onlytwoofthe**alleles**canbepresentinasingleorganism. For example,theABOsystemofblood groups is controlled by three **alleles**, only two of which are present in an individual.

Genetic Disorder?

A **genetic disorder** is a **genetic** problem caused by one or more abnormalities in the genome. Most **genetic disorders**are quite rare and affect one person in every several thousands or millions. **Genetic disorders** may be hereditary, meaning that they are passed down from the parents' **genes**.

Genetic Mapping?

Gene mapping describes the methods used to identify the locus of a gene and the distances between genes. The essence of all genome mapping is to place a collection of molecular markers onto their respective positions on the genome.

What Is Location Of Intronic Sequence With Respect To Exons?

Sequences that are joined together in the final mature RNA after RNA splicing are**exons**.**Introns**are**found**inthegenesofmostorganismsandmanyviruses, andcanbe**located**inawide rangeofgenes,includingthosethatgenerateproteins, ribosomal RNA(rRNA), andtransfer RNA(tRNA).

Write Down At Least Three Mitochondrial Diseases?

MELAS

Mitochondrial encephalomyopathy, lactic acidosis and stroke-like episodes is MELAS. Kearns-Sayre Syndrome

NARP

NARP stands for neurogenic weakness, ataxia and retinitis pigmentosa.

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Structul chamge in chromosome?

Structural Change in the **Structure** of **Chromosomes .....Chromosomes** are the vehicle of hereditary material or genes. Any alteration, addition or deletion of **chromosomal** part leads to alteration of number, position or sequence of genes in the **chromosome**.

•Define Epistasis

Epistasis and modifier genes Epistasis is a phenomenon that consists of the effect of one gene being dependent on the presence of one or more modifier genes Type of polygenic inheritance where alleles at one gene locus can hide or prevent expression of alleles at second gene locus.

•Genomic Imprinting

Genomic Imprinting Epigenetic marking of a locus on the basis of parental origin. Results in monoallelic gene expression. Some genes are expressed only from the maternal genome and some only from the paternal genome. It is estimated that about 40 genes are imprinted and they can be found on several different chromosomes .

Point mutations

- Point mutations affect single sites on DNA.
- Substitution of one base for another.
- Deletion/addition of a single base or small number of bases.

Transitions

- If purines (A/G) is replaced by purines and pyrimidines (T/C) is replaced by pyrimidines.

Transversion

- If purine substitutes for pyrimidines or pyrimidine is replaced by purines. It is called as transversions.

Result of point mutations

- Silent mutations
- Missense mutations
- Nonsense mutations

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—

- Frameshift mutations
  result in all amino acids downstream from the mutation site being completely different from wild type. These proteins are generally nonfunctional.

Mutation - Forward, Reverse and Neutral Forward mutation

- Mutation that changes the wild type phenotype into mutant type phenotype.

- Any mutation which changes a functional wild genotype or wild phenotype to nonfunctional form.

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Wild type phenotype/ genotype

— **reverse mutation**	Most common phenotype in natural populations of the organism is called a wild type phenotype.
— **neutral mutation**	Mutation that changes a mutant phenotype back into the wild type.
—	Mutation that alters the amino acid sequence of the protein but does not change its function as replaced amino acid is chemically similar or the affected amino acid has little influence on protein function.

Common X-linked dominant diseases

- Rett syndrome
- Alport syndrome
- Goltz syndrome

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- Fragile-X Syndrome

Short Tandem Repeat Polymorphism

Short tandem repeat sequences

- STR are repeats of nucleotide sequences.
- AAAAAA… mononucleotide
- ATATAT… dinucleotide
- TAGTAGTAG…trinucleotide
- TAGTTAGTTAGT… tetranucleotide
- TAGGCTAGGCTAGGC… pentanucleotide
- Different alleles contain different numbers of repeats.

Frameshift mutations

result in all amino acids downstream from the mutation site being completely different from wild type. These proteins are generally nonfunctional.

Common mitochondrial diseases

- Myoclonic epilepsy
- Mitochondrial recessive ataxia
- Leber hereditary optic neuropathy
- Sensory ataxia
  neuropathy 17.

Chiasmata formation —

Chiasmata visible on chromosomes are regions of crossing over. — Occurs between

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non-sister chromatids

Crossing over

is the swapping of genetic material that occurs in the germ line. During the formation of egg and sperm cells, also known as meiosis, paired chromosomes from each parent align so that similar DNA sequences from the paired chromosomes cross **over** one another.

During meiosis I, homologous chromosomes separate. During meiosis II, sister chromatids separate.

Prophase I

— Nucleolus disappears. — Meiotic spindle forms. — Nuclear membrane disappears.

Spermatogenesis and oogenesis

Meiosis in males is called spermatogenesis and produces sperms.
Meiosis in females is called oogenesis and produces ova.

— Forward genetics

Not knowing anything about the location of the gene.
— Starting at the phenotype, determining which protein was involved and then getting to the gene through the protein.

reverse genetics.

Starting from the approximate location of the gene, to finding the gene itself.
— Translating it to learn about the protein and its function
Positional cloning or reverse genetics.

nondisjunction chromosome

Nondisjunction occurs when homologous chromosomes (**meiosis** I) or **sister chromatids** (**meiosis II**) fail to separate during **meiosis**

Types of Non-disjunction

Meiotic non-disjunction
Mitotic non-disjunction

Meiotic Non-disjunction If non-disjunction occurs during 1st meiotic division then all the sperms/ova derived from primary spermatocyte/oocytes will be abnormal.

Mitotic Non-disjunction Mitotic non-disjunction results in cells with different number of chromosomes. Breaking of the spindle fibers during metaphase or anaphase.If spindle fibers do not form for one of the chromosomes, then during anaphase it will not separate into two chromatids and one cell will have an extra chromatid and the other will be lacking one. Mitotic nondisjunction causes trisomic and monosomic daughter cells to be formed.

SNP: Single Nucleotide Polymorphisms

Haplotype mapping: sets of alleles or DNA sequences can be clustered so that a single SNP can identify many linked SNPs Single nucleotide polymorphisms (SNPs) are

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polymorphisms that are caused by point mutations that give rise to different alleles containing alternative bases at a given position of nucleotide within a locus. Due to theirhigh abundance in the genome, SNPs already serve as the predominant marker type.

Monosomy:

Monosomy is the state of having a single copy of a chromosome pair instead of the usual two copies found in diploid Cells Causes of monosomy: followings are the causes of monosomy

Non disjunction – one gamete receives two copies of homologous chromosomes and other will′ have no copy.

Loss of chromosome. As it move towards pole of cell during anaphase. Fruit fly′ **monosomy**, is the cause of some human diseases such as Turner syndrome and Cri du Chat syndrome

What are structural abnormalities?

Structural abnormalities are when part of an individual chromosome is missing, extra, switched to another chromosome, or turned upside down.

Types based on genetic material

Balanced – chromosome complement is complete.′

Unbalanced – when there is incorrect amount of genetic material.′

Types

Translocations′ Deletions′ Insertions′ Inversions′ Ring chromosome′ Isochromosome′

Parents of affected individuals are asymptomatic

In most of the cases, parents of affected individuals are asymptomatic.

Deletion

For point **deletions**, one nucleotide has been deleted from the sequence.

cystic fibrosis, Turner syndrome,

Results in frameshift mutation.

Suppressor mutation

Suppresses the effect of other mutation

Occurs at a site different from the site of original mutation.

Intragenic suppression

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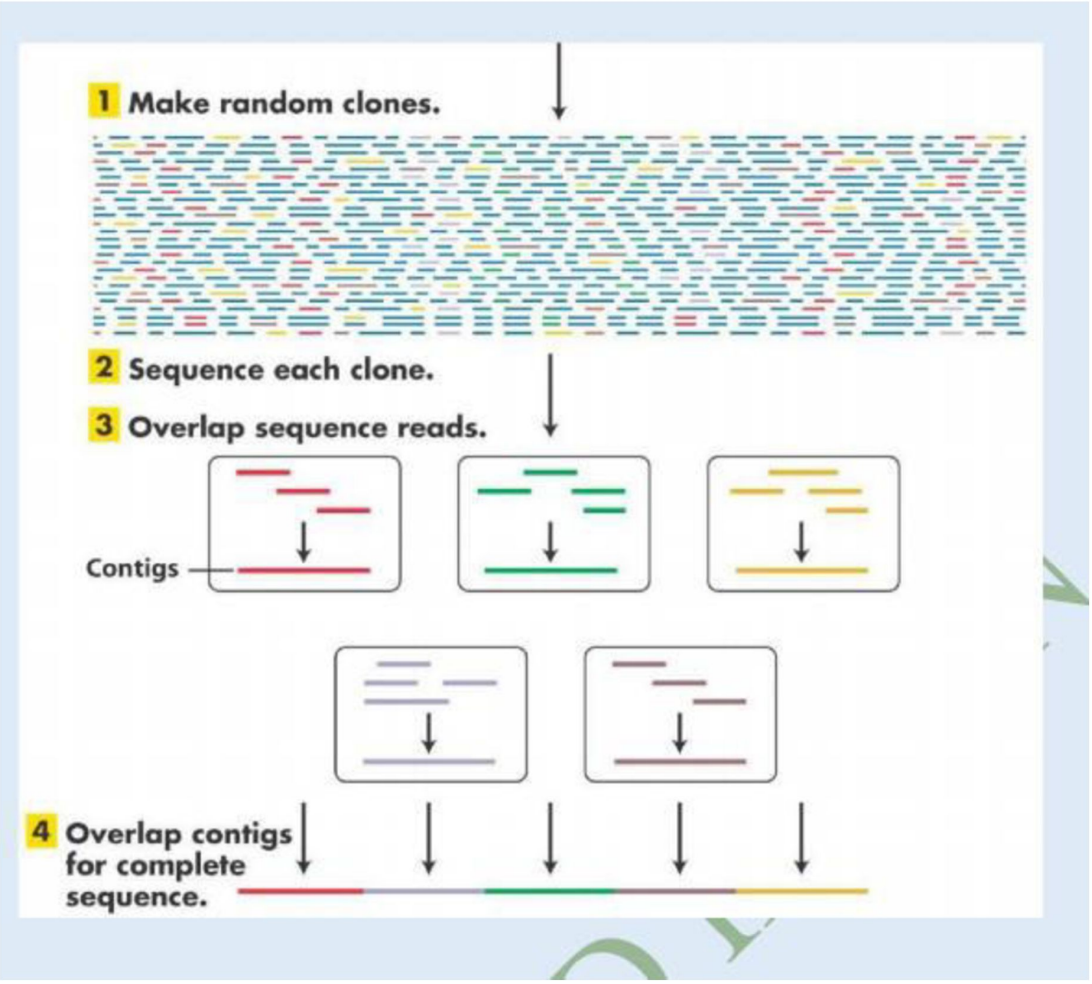
Intragenic suppression results from suppressor mutations that occur in the same gene as the original mutation.

Intergenic suppression

Intergenic (extragenic) suppression relieves the effects of a mutation in one gene by a mutation somewhere else within the genome. Second mutation is not on same gene as original mutation

Bottom-up Mapping



Penetrance and Expressivity

Penetrance measures the proportion of individuals in a population who carry a specific gene and express the related

trait. Less than 100% penetrance is a result of modifiers, epistasis, suppressors, and environmental conditions.

In **genetics**, **expressivity** is the degree to which a phenotype is expressed by individuals having a particular genotype. (Alternately, it may **refer to** the expression of particular gene by individuals having a certain phenotype.)

Low and full Penetrant

Low Penetrance An allele with low penetrance will only sometimes produce the symptom or trait with which it has been associated at a detectable level. In low penetrance, it is difficult to distinguish environmental from genetic factors.

Highly Penetrant If an allele is highly penetrant, and then the trait it produces will almost always be apparent in an individual carrying the allele.

Polyploidy :

Polyploidy is the heritable condition of possessing more than two complete sets of chromosomes Two major types

Autopolyploidy♠

Allopolyploidy♠

Frequency of autosomal dominant disorders

Frequency of autosomal dominant disorders varies between 3 – 9.5 per 1000 individuals

Frequency of autosomal recessive disorders

Frequency of autosomal recessive disorders is between 2-2.5 per 1000 individuals.

Effect DNA sequence but not protein: Silent mutation. Mcq and definations

Trisomy

Trisomy is a type of polysomy in which there are three copies of a particular chromosome, instead of the normal two.

Trisomy is a type of aneuploidy. Trisomy of chromosome 21, which is found in Down syndrome, is called trisomy 21

inversion mutation

Inversion mutations only affects a small part of the gene.

 May a single nucleotide or more than one nucleotides are involved.

X-linked recessive disorders

Duchene Muscular Dystrophy

Hunter's Disease

Menkes Disease

Hemophilia A and B

Color Blindness

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Diagnostic tests for genetic disorders

Physical analysis: Physical analysis for affected individual

Ultrasonography of fetus: Ultrasonography of fetus determines malformations of head, internal organs, and extremities

Analysis of amniotic fluids: Amniotic fluid analysis to determine genetic and chromosomal disorders after 14 wks gestation. More than 200 various genetic diseases easily be detected.

Maternal blood analysis: Maternal blood analysis to observe abnormal fetus.,

Polymorphism

,

Polymorphism, in biology, a discontinuous genetic variation resulting in the occurrence of several different forms or types of individuals among the members of a single species

Aneuploidy: Change in chromosomes number can occur by the addition/deletion of chromosome or part of a chromosome

. Euploidy: Gain of one or more complete sets of chromosomes

pattern of inheritance

Most genes have one or more versions due to mutations or polymorphisms referred to as alleles. ... There are five basic modes of **inheritance** for single-gene diseases: autosomal dominant, autosomal recessive, X-linked dominant, X-linked recessive, and mitochondrial.

Mutation that produce the change in the amino acid in protein product (histidine in for arginine Missense mutation

mcq and definition

Multifactorial diseases six

: Genetic disorders controlled by many genes plus the effects of the environment

Asthma,Cancers ,Diabetes ,Heart diseases,Hypertension,Obesity

Pleiotropy

Pleiotropy The ability of a gene to affect an organism in many ways is called pleiotropy.

 Pleiotropy refers to an allele which has more than one effect on the phenotype.

Types of deletion include the following :

Terminal deletion – a deletion that occurs towards the end of a chromosome.

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Intercalary/interstitial deletion – a deletion that occurs from the interior of a chromosome.

Microdeletion – a relatively small amount of deletion (up to 5Mb that could include a dozen genes).

Applications of molecular markers – short range

- Parentage determination
- Genetic distance estimation
- Determination of twin zygosity

Applications of Molecular Markers – Long Range

- Gene mapping & mapping of QTL by linkage
- Markers assisted selection of breeds

Method of mapping gene disorders

Genetic maps have been used successfully to find the single gene responsible inherited disorders, like cystic fibrosis and muscular dystrophy.

How genetic mapping performed

Scientists isolate DNA from samples and examine it for the unique patterns of bases.

Molecular techniques are referred to as genotyping polymorphisms or markers.

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MID TERM QUIZE

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1. Primary oocyte produces polar bodies 2
2. Which two alleles are called co-dominant one another and are expressed when present?__ _____-__ Allele A and Allele i
3. During recessive epistasis ,which color of pigeon is produced by recessive red gene with orwithout spread?- _____Any color
4. The DNA between nucleosome is called as linker DNA .Its length is ..... (d) 20-60 bp
5. Mendel determined the segregation of alleles at single gene locus basedon .....Monohybrid
6. Darker bands are generally found near the centromere or on the ends (telomeres) ofchromosomes.

   The position of the dark –staining are .....or.....Heterochromatic region ,heterochromatin

7:During metaphase mitosis chromosom_____Line up at the equator

8. During stage telophase 1 ,chromosomes are....._____Haploid

9:The law of independent assortment with dihybrid cross gives genotypic ratio in F2 as:___ 9:3:3:1

10:Who is considered "The Father of Genetics"?_____Gregor Johann Mendel

11:In .....centromere splits into two parts while in .....it does not split.__Mitosis,Meiosis I

12:When one phenotypic trait is determined by multiple alleles then it is called as .....Pleiotropic

14:Which is hereditary disorder characterized by lung congestion ,infection,and malabsorption of nutrients by the pancreas_____Huntington disease

15:which one of the following histone protein has molecular weight ,greater number of amino acidand high percentage of lysine and arginine?_____H1

16 is an example of inherited variation?_____ Finger prints

17:When chromosomes stained they show a dark –stained region that is ..... Centromere

18:Histones have two basic amino acids _____Arginine-lysine

19:The diameter of DNA nuclosome on the chromosome scaffold is ?_____11nm

20. e coli has about,..... Genes organized into its one cirularr chromosome_____3000 to 4000.

21:Which is the following blood group is considered a universal donor?_____O

22:Which type of genetics deals with movement of genes and genetic traits from parents to offspring?

)Transmission genetic

23:Which statement is correct about genetic and physical mapping?

The some genes are close together on the genetic map and are also close together on physical mapof the chromosome

24:Polygenic traits are also affected by the persons environment .so they are called__?

25:Which is the histone deacetylase inhibitor?_____vorinostat

25:Generally,karyotype represent by arranging in the .....order of size keeping their centromerein .....?

Descending ,straight line

26:The spindle apparatus is formed during the .....phase of mitosis_____Prophase

27:During .....the effect of prophase and pro metaphaseare reversed.____Telophase

28:Meiosis result in..... _____4 Haploid daughter cells

29:The allelic series refer to_____Multiple alleles

30:Which of the following statement is correct about genes(a)30% are mono allelic and they exist in two forms__ _____70% are mono allelic and they exist in one form

31:The crossing over occurs a_____Chiasmata

32:The longest stage in the cell cycle is_____Interphase

33:Which of the following statement is not correct about continuous variation?

Continuous variation controlled by . a lot of genes

35:Mutational changes are .....and epigenetic changes can be.....Irreversible,reversible

36:In .....sister chromatids separate and move along the kinetochore microtubules toward oppositeends of the cell._____Anaphase

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37:Position dependent method of diseased gene identification follows

Reverse genetics

38:Which statement is not correct about epigenetic regulation in identical twins? The epigenetic tags can have such ___ an effect on the twins that one can develop a disease and the other is also

39. The difference in the twins epigenomes is what makes them become different when they are older

The epigenetic tags can have such an effect on the twins that one can develop a disease and theother is also

40:Which one of the following statement is correct?

Father with X-linked disorder will transmit the mutant allele to all daughters but to no sons

41:Band that indicates constitutive heterochromatin……_____band

42:What is the process of transfer of human blood known as?_____Transfusion

43:All offspring of same variety is called a_____True breed

44:In DNA double helix,the two DNA chains are held together_-_ Hydrogen bonds between the pair of bases

45:If 14% of the nucleotides from a DNA molecules contain the base T.what percent will contain thebase G?_____14

46 are proteins used to package DNA_____Histone

47:Darker bands are generally found near the centromere or on the ends(telomeres)of the chromosome.The position of the dark staining a_____Heterochromatin region ,heterochromatin90'

48: According to genetic material of parents mixes during fertilization.__ ___Blending hypothesis

50:On what basis four major blood groups is classified as positive and negative blood type?

Rh factor

51:Which one of the following statement is correct about to the Mendel Law of Dominance?

In a cross of parents that are pure for contrasting traits only one form of the trait will appear in thenext generation

52:Chromatin fibres undergo folding ,coiling and supercoiling during so that the chromosomes become progressively thicker and smaller._____Prophase

53:Genetcs is the study of_____All

54:Which one is building blocks of chromosomes?_____Nucleosome

55. Genetists or researchers can study the genotype based on the phenotype in

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classical_____DNA sequencing

56:During the summer,the arctic fox produces enzymes that cause its fur to become reddish brown .During the winter ,these enzymes donot function .as a result the fox has a white coat that blends into the snowy back ground .This change in fur colors shows that_____ _____The expression of certain genes is affected by temperature

57:According to particulate hypothesis ,parents pass on distinct heritable units .Those distinctheritabl units are now c

59:During stage telophase I,chromosomes are……..(a)Haploid

60:Mendel determined the segregation of alleles at single gene locus based on_____ _____Monohybrid cross

61:How much length of the DNA sequence within kinetochores regions?_____120 bp

62:What is the unit of linkage map?_____morgan

63:which one of the following statement is not correct about Co-dominance?

64:Prophas_____Chromosomes condense and remain attache

65:Due to during meiosis I,the two sister chromatids of each chrosome are no longer genetically identical_____Crossing over

66:DNA replication is a type of …._____Semi-conservative

67:Which one of the following statement about prokaryotic chromosomes is wrong?___ ____Prokaryotic chromosomes made of DNA and Histone protein

68:70%genes in human are_____Mono allelic

70:Identify the purine base of nucleic acids in the following?_____Adenine

71:The diameter of packed nucleosome is_____ 30nm

72:In DNA double helix ,the two DNA chains are held together by_____ _____Hydrogen bonds between the pair of bases

73:Which one is not regulate gene expression?_____Nucleotide medication

74:Epistasis is the interaction between ……genes.)_____-2

75:Technique that gives an rough idea of gene location_____?

76:A female bee sex is_____ZW

77:Fragile X chromosomes affect the_____FMR1

78:Stage of prophase at which nuclear membrane dissolves…._____Stage 2

79:The law of independent assortment with dihybrid cross gives genotypic ratio in F2 as._____9:3:3:1

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80:Who is the father of genetics?

81:Which of the following is not the property of garden pea plant that was used by Gregor Mendel forhis experiment?_____Long generation time

82:How many numbers of alleles control the fur color of rabbit?_____Four

83:The ABO blood group in humans is determined by_____Multiple alleles

84:The giant chromosomes found in …… and they may be as long as 300 micro and 10 micro indiameter.__ _____Diptera

85:Which map is primary based on the location of landmarks along a DNA molecule?-_ _____physical mappin

86:Genetic make up of an organism is called his experiment?_____genotype

87:The type of cell division important for tissue regeneration is_____mitosis

88:Chromosomes were described by ……in 1875 and chromosome was first used …..in 1_____ _____Strausberge,waldeyer

89:During prophase ,the nuclear envelope……_____ Disappear

90:……..occurs when one mutation has the opposite effect when in the presence of another mutatin _____Sign epistasis

91:which of the following has longer chromosomes…..._____ Plants

92:A heredity unit chromosomes is called…_____ Genotype

93:Which one of the following is not correct about the influence of environmental factors on.. epigenetics?_____Social interactions and physical activity not affect the epigenome

94:which of the following statement is correct?_____Father with X-linked disorder will transmit the mutant allele to all daughters but to no son

95:Name the organism whose first genetic map was made?_____Drosophila

96:Which one the statement is not correct about the position dependent method of disease causinggene ____Starting at the phenotype ,determining which protein was involved and then getting through theprote

97:Which is the correct sequence of bottom up mapping technique?_____make random clones—sequence each clone----overlap sequence reads over lap contigs for complete sequence

98:The numbers of genotypes can be calculated on the basis of ……._____Number of alleles

99:When a cell divides.what do we call the 2 new cells_____daughter cells

100:which statement is not correct about recessive epistasis in pigeons?_____Patteren is epistatics to spread color

102:chromosomal theory of inheritance states that behavior of chromosome is according to___ Both

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103:How many hours does the G1 phase of interphase take to complete a cycle?_____ 6-12 hours

104:In physical map,units of distance are expressed in_____Centimorgan or cM

105:Which of the following statement is not correct about Meiosis I?_____Meiosis I results in two haploid daughter cell

106:A nucleosome consists of)_____DNA and Histones

107:Which type of inheritance appears in AB blood group phenotype?_____Co-dominance
108:The DNA in a bacterial chromosomes is best described as:_____ A single circular double –helical molecule
109:In humans ,a zygote is a ……..cell having …….chromosome_____ Diploid,46
110:Which trait will produce in F1 generation of the crossing a brown pepper (RRyy) with a yellow pepper(rrYY)_____Brown
111:Which one of the following is an example of diploid cells?_____All
112:The splitting of sister chromati

113:Which are major factors that affect the epigenome?_____Expoure to toxins and diet

114:Which one of the fur color of rabbit is controlled by maximum numbers of genotype___Gray

115:Spindle microtubules attach to the kinetochores of chromosomes and move the chromosome tothe_____ Metaphase plate

116:There are about …….loops in the chromosome of E.choli .These loops are highly twisted orsupercoiled . structure with about …..mlilion nucleotide pairs._____50,

117:How many most common chromosomal banding types are present?_____4

118:Giant chromosomes are first discovered by ……in 1881 in dipteran salivary glands

119:Pepper color is controlled by_____Two gene
120:During anaphase of mitosis push against each ,that stretches the cell into an oval shape._____Non –kinetochore spindle fibers

121 provide terminal stability to the chromosomes and ensure its survival._____Telomeres

128:Differential expression of the genetic material depending on its parentage of inheritance gives……_____Expressivity

129:Which of the following is not the part of interphase in the cell cycle?_____ M phase

130:Nuclear DNA replicates in the …… phase._____s phase

131:Which type of inheritance appears in AB blood group phenotype?_____ Co-dominance

132:Centromere breaks by the removal of protein ….._____ Tubulin

133:Historically ,firstly developed sub discipline of genetics is…._____Transmission genetics

134:Which of the following statement is correct?_____Carrier female Mates normal male,50% chance that each daughter will be carrier,50% chancethat each son will have the / . .disorder

135:In the parental generation,when Mendel crossed a true breeding pea strain that produced

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smooth seeds with one that produced wrinkled seeds then result was….._____All the F1 progeny seeds were smooth

136:Which one of the statement is not correct about the position independent method of disease causing gene mapping in human ?_____Starting at the phenotype , determining which protein was involved and then getting to the gene through the protein

137:Which of following blood groups are consists of two different genotypes as homozygous and heterozygous?_____A&B

138:Differential expression of the genetic material depending on its parentage of inheritance gives…_____ Expressivity

139:The molecular weight of the histoneprotein H4 is_____21130

140:How many numbers of alleles control the fur color of rabbit?_____four

(141:…….occurs when two deleterious genes are beneficial when together ___ _____ Reciprocal epistasis

142:In which of the cats the pattern observed in coat color is due to environmental effect?_____Siamese \

143:The sugar molecule in a nucleotide is_____Pentose

144:Which one of the following is show the pleiotropic effects in humans?_____ All

145:When one phenotypic trait is determined by multiple alleles then it is called _ Polygenic inheritance

146:Which of the following is not correct when gene is switched OFF and transcription is not possibl_____ Acetylated histones

148:Histone modification are cause of_____Epigenetics

149:More than ……specific trinucleotide repeats is seen in individuals with fragile 200

150:Fruit color of summer sqush plant is an example of_____Dominant epistasis

151:How many pairs of homologous chromosomes are present in human cell?_____22

153:In life forms with the ZZ-ZW chromosomal sex determination ,the females are ome not visible in Homogametic

154:Which one restores the paired condition for both chromosomes and genes_ Fertilization

… ….due to ……but c155:Which of the following organism has highest numbers of chromosomes ?_____Horsetail

156:Which of the following have single genotype_____ B&O

157:Meiosis,unlike mitosis involves_____1 nuclear division

158:which of the following is not property of Garden pea plant that was used by Gregor Mendel for his experiment_____Long generation tim

159:The mitotic apparatus microtubules comprises can be seen_____ All

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160:……variation describes the situation in which there are many intermediates between theextremes._____Continuous

162:Which one of the following statement is correct about human genes?_____30% genes are di allelic and in two forms

163:The molecular weight of histone protein H1 is_____ 22500

164:Genetic material in cells is organized into chromosomes ,prokaryotes generallyhave……chromosomes.eukaryotes have ….chromosomes in their nuclei._____One circular ,multiple and linea

165:Which epistasis between beneficial mutation generates improvements in function_ \ _____*Positive epistasis*

166:Exchange of DNA segments between chromosomes is known _____Crossing over ngiosperm-78

167:Which one of the following is not correct about organism and with their chromosomes number ?

Positive epistasis

168:The frequency of crossing over between any two linked genes is_____Proportional to distance

169:A linkage map_____Both

170 :G0 phase is '……__post mitotic phas

171:The cytokinesis in animal cells is characterized by_____Cleavage furrow

172:Which of the following has longer chromosomes…..+_____Plants

173:Which one of the following genotype expresses the green_____rr|yy

174:When DNA polymerase is in contact with guanine in the parental strand ,what does it add to thegrowing daughter strand?_____ Cytosine

175:When two mutations together have a smaller effect than expected from their effects when alone,it is called_____ Negative epistass

176:Which one of the following have chromosomes number from 8-16 and most common number is .
177_____Angiosperms

177:In case of two gene interaction .the gene which is making the expression of another is called……_____Dominant

178:DNA replication is type of …..____Semi-conservative

179:Hair color is controlled by alleles on …Chromosomes 3,6,10,18

180:Groups of human chromosomes are…..____ 7

181:During the summer ,the arctic fox produces enzymes that cause its fur to become reddish

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brown,During the winter ,these enzymes do not function .As a result,the fox has white c oat that blends into the snowy background.this change in fur color .n showthat_____The expression of certain genes is affected by temperature

182:All offspring of same variety is called as_____True breed

183:Which one of the following statement is not correct about incomplete dominance?_____ One of the allele is dominant

184:Stage of prophase at which nuclear membrane dissolve…_____ Stage 2

185:In meta phase I,……line up at metaphase plate with one chromosome facing on ……*Tetrads*

186:Human chromosomes are_____Both

187:The diameter of chromatin i_____30nm

188:The life cycle consists on_____All

189:Building block of nucleic acids are…..._____ Nucleotide

190:……,states that the two alleles of gene found on each of pair of chromosomes separate independently of one another into sex cell_____Independent assortment

191:E.choli has about genes are organized into its one circular chromosome

_ _____3000-4000

192:The genome length of prokaryotic chromosome is smallest in

RNA virus

193:Which one of the following statement is correct about variation?____ Both

194:Spindle microtubules attach to the kinetochores of chromosomes and move the chromosomes to .. . . the_____Metaphase plate

195:Modern concept about the law of dominance is……_____The gene for a particular inherited character resides at specific locus homologous chromosome

196:Technique that gives an rough idea of gene location_____genetic mapping

197:In blue_eyed Mary plant a white substance is first converted to substance by 1st gene and that to a ……substance by 2nd gene to give a blue flower_____colorless,magenta

198:……..variation describes the situation in which there are many intermediates between the extreme_____continous

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extreme

199:……..cells become …….cells of the embryo which differentiate into specific lineage_____

_____ . totipotent,pluripotetn

200:Which condition is called as polygenic inheritance?_____ Two genes influence one trait

201:How many number of bands can be recognized on giant chromosomes of the Drosophila?_

_____5000

202:What is the other name of polytene chromosomes?_____Salivary gland chromosomes

203:In the ABO system,blood group O is characterized by the_____ Absence of both antigen A and antigen B

205:what is the name of the region where double stranded singular circular DNA is found in the prokaryotic cell?_____ Nucleoid

206:What is the composition of nucleotides?_____-A base +a sugar +phospholipids

207:Which of the following is widest DNA?_____A-DNA

208:Causes genes to not segregate independently ,causes genes to be more likely to be inherited

208:Causes genes to not segregate independently ,causes genes to be more likely to be inherited Together_____gene linkage

209. A DNA polymorphism is …..of population_____1-2%

210:Tetrads align at center …._____ Metaphase I

211:Meiosis in males is called ……and produces sperms_____Spermatogenesisz

212:Primary occyte produces polar bodies…._____ 3

213:In humans ……of somatic cells_____ ~220 types

214:Each identical side of a chromosomes is the_____Chromatid

215 : How many types of enzymes are involved in histone acetylation?.....2

216:The region where two sister chromatids of a chromosome joined or held together during mitoiticmetaphase is called…_____Centromere

217:Which trait will produce inF1 generation of the crossing over a brown pepper (RRyy) with yellow pepper . (rrYY)?_____Red/

218:In life forms with the ZZ-ZW chromosomal sex determination ,the males have__

_____ZZ chromosomes

219:which one of the following statement is correct about to the Mendel Law Of Dominance ?

In a cross of parents that are pure for contrasting traits only one form of the trait will . appear in the next generation

220:A point mutation in the …….completely eliminates the ability of the centromere to function during chromosome segregation_____ CDE- III sub domain of CEN DNA

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221:Which one of the following statement is not correct about Prophase II?__

___- Replication of chromosomes occur

222:Which statement is not correct about epigenetic regulation in identical twins?_____

_____The epigenetic tags can have such an effect on the twins that one can develop a disease andother is also

223:Which one of following organisms have 16 chromosomes ?_____Onion and mold

224:T.H Morgan test cross …..Crossing over and linkage genes that breaks the gene physical connection

225:Which of following does not revert to the euchromatin stage?_____Constitutive heterochromatin

226:Which one of following statement about color blindness is correct?

Color blind mother will transmit the mutant alleles to all sons but to no daughter

227:Which of following statement is correct about the epigenetic modif__ When histones are acetylated chromatin is open and genes are potentially active

228:How many numbers of heterozygotes genotypes will be produced from five number of genespecific alleles?_____ 15

229:Genes will be switched ON when_____Histones are acetylated

230:Which one is first time observed by Flemming in 1882?_____ Lampbrush chromosome

231:Chargaffs rules for DNA base pairing indicates that_____-A+T=G+C

232:Failure in which of the following phenomenon would result in linkage?

234:Skin color pigmentation in humans is controlled by at least …..separately inherited genes?__ 3

235:Formation of pluripotent cells from totipotent cells is called_____ Cellular differentiation

236:What type of segregation occurs when alleles at the two gene loci segregate together and are .transmitted unit Dependent segregation

237:Fragile X syndromes is most commonly caused by

238:Which one of CG_____CGG

239. the following statement is not correct about homologous chromosomes_____ Similar in size

240:How many nusmbers of alleles can generates 3 genotypes?_____2

241:Group of adjacent nucleotides are joined by_____Phosphodiester bond

242:Which is correct about the comb shape in chickens?_____Four different comb

243:…….occurs when one mutation has the opposite effect when in the presence of another mutation.____ _____ Sign epistasis

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244:A cell with 10 chromosomes undergoes mitosis .How many daughter cells are created? Each daughter cell has chromosomes_____2,10

246:which one is not regulate gene expression?_____Nucleotide modification

247:During recessive epistasis ,……,gene exerts its effectwith …..genotype_____Epistatic ,homozygous,recessive

248:Name of the organisms whose first genetic map was made?_____Drosophila

249:histone modification is cause of_____ Epigenetics

250:Which statement is not correct about recessive epistasis in pigeons?_____ Recessive red

251:Haploid gametes are formed by

252:How many numbers of genes are present in yeast?_____5800

253: Nuclear DNA replicates in the ……phase._____S phase

254:The type of cell division important for growth is_____Mitosis

255:Kinetochore are formed at_____ Centromeres

256:which stage of the cell cycle are arrested before staining?_____Prophase

257:which type of genetics deals with the movement of genes traits_____Transmission

259:The pleiotropic effect is shown in human_____All

260:The molecular weight of histone protein is_____ 22500

261: The diameter of DNA double helix is_____2nm

262:In which of the cats the pattern observed in coat color is due to environmental effect?___ _____Siamese

263:According to …….genetic material of the parents mixes during fertilization__ _____ Blending hypothesis

265:when a cell divides ,what we call 2 new cells?_____Daughter cells

266:A DNA polymorphism is …..of population_____ 1-2%

267:Generally ,karyotypes represent by arranging the order of size keeping their centromere in a …._____ Descending ,straight line

268: when ice pack is attach to Himalayas rabbit …._____Color will be darker

269:What is phosphodiesters linkage_____ Both

271.----------Which of the following is true about B-DNA?
planes of the base pairs nearly perpendicular to the helix

[

272. -------Which one of the lowing statement is conect about X-chromosome-tvstion phenomena?
One expression on one of the male X-chromoson is down regulated due to DNA methyletion an O

273. -----Darker bands are generally found near the centromere or on the ends (telomeres) of

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Hetrochomatic region heterochromatin

274. ----------Genetics is the study of
All of the mentioned

275. ---------- Wachu dicipline is called on Mondulien genetics?
Quantitative genetics

276. ----------: Epestesis is the interaction between

277. ----------Exchange of DNA segments between chromosomes is known as_____Chiasmata

278---------: Which one of the following statement about eukaryotic chromosome is wrong?
Eukaryotic chromosomes found in a cytoplas

279. ----------The correct sequence of Meices I prophans stages is?
Leptotene, zygotene pachytore diplotne danes

280. ---------- Which of the following enzymes remove helix and produces sick to
Dyrses and Topoisome

281. ----------10-15% of chromosomal deletions occur due to
5000

282. ----------Chromosomes having centromere at one end of arms
Telocentric

283. ====== The agents which induce artificial mutations are called
Mutagenic agent

284. ----------In case of summer squash, which statement is not correct?
Dominant Y locus produces yellow color

285. ----------During ..,.....push against each other, that
Non ko.....

286. ----------Which one of following allele controls the Chinchilla (ight gray) color of the robbit?c
_____C

287. ----------The albitic series refers to_____Lethal alickes

288. ----------: Inherited variations genetically .....and cannot be altered
Controlled

289. ----------: What enzyme is responsible for copying methyletice marks fces the
per tot of DNA during replication?_____DnmTg

290----------: .....Causes a trait to appear in inappropriate tissues or at inappropriate
times during the process ofdevelopment_____Grain of function

291. ----------: In physical map. units of distance are expressed_____Centmorgan of Cm

292. ---------- The diameter of DNA double helix _____11nm

293. ----------Which one of the following genotype expresses the Green color of pepper?__rrlyy

294. ----------: A nucleosome consists))_____Dna and histones

295. ----------: Which one of the following statement is not correct about Co-dominance?
Both alleles are expressed in a heterozygous individuals

296.. ----------Which of the following is not correct when gene is switched OFF and
transcription is not_____ Acteylated histones

297. ----------: E coli has about .....genes are organized into its one circular chromosome
_____3000to 4000

298. ----------Which one of following statement is not correct about Homologous
Chromosom Pair of chromosomes that are similar in shape

299. ……Which of the following is not a common mode of histone modification in eukaryotes?
Sulphonation

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300. ----------Band that indicates constitutive heterochromatin_____G band
301. ----------A female bee sex is_____Dipkid
302. ----------: Most common phenotype in natural populations of the organism is called
Wild type phenotype
303. ----------: How many numbers of heterozygotes genotypes will be produced from
five numbers of _____Ten
304. ----------In Me forms with the ZZ-ZW chromosomal sex determination, the males are
Homogametic
305. ----------: Which of the following is true about B-DNA?
Planes of the base pairs nearly perpendicular to the hela exis
306. ----------Which of following statement is correct about the epigenetic modification?
When histones not tagged deacetylated the chromatin open and genes active
307. ----------Which one of following statement about color blindness is correct?
Color blind mother will transmit the mutant allele to 50% sons but to 50% daughters
308. ---……..alters the amino acid sequence of the protein but does not change to function
replaced _____amino acid is chemically similor or the affected amino and has lelo luence
on pi function_____Neutral mutation
309. ----------: The type of cell division important for growth is_____Mitosis
310 . ----------Suppose that the distance between two genes on a linkage mep is 7 units.
The crom over frequency _between the two genes is__-_____7%
311. ----------: Which one of the following is not correct about organism with their
chromosome numbers? _____Angiosperms have
312. ----------: Which one of following statement is not comect about Mis
Meiosis I results in two haploid daughter cells
313. ----------: Name the phenomenon which can break the occurrence of linkage?
Crossing over
314. ----------: During metaphase mitosis chromosome
Lineup at the equator
315. ----------Hetorically, firstly developed sub discipline of genetics is
Transmission genetics
316. ----------Which statement is not correct about epigenetic regulation in identical twins?
The epigenetic tags can have such an effect on the twins that one can develop a
disease and the other is also
318. ----------: How many hours does the G1 phase of interphase take to complete a cycle?
6-12hours
319. ----------: How many numbers of alleles can generale 10 genotypes?
Genes with 4 alletes
320. ----------: In ife forms with the ZZ-ZW chromosomal sex determination, the males have
zz checmasone
321. ----------: Chargaff's rules for DNA base pairing indicates that
A-T and G-C
322. ----------Dicots in general, have a.... number of chromosome than monocots.
. Chromosomes a..... in monocot than dicots_____Higher long

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323. -----Mitotic apparatus helps_____Chromosome inheritance
324. ------Which of the following produces monosomic and trisomic daughter cells Select the Meiotic I non-disjunction
325. --------[: When chromosomes stained they show a dark-stained region that is the Centromere
326. ----------What will be dihybrid test cross ratio if genes are located on different chromo _____933
327. -------- Which one of following organisms' series have no metacentric autosomal chromosome Dog.goat.cow
328. ---------- Historically, firstly developed sub discipline of genetics is___Transmission genetics
329. ----------What is the division of cytoplasm colled_____ Telophane
330. ----------: During anaphoon of loss push against each other that stretches the call Non kinetochore sprdietbers
331. ---------- Which of the following blood group es considered a universal donor?___B
332. ---------- Which of the following is not correct when gene is switched ON and transcription is possib _____Acetyloted histone
333. ---------- The region where two sister chromatids of a chromosome joined or "held together" Parms
334. ---------- Fegle X chromosome affects the gene_____Fmr1
335. : - ----------How many numbers of homozygotes genotypes will be produced from_4
336 . ----------: In which of the following fur color of the rabbit is controlled by Chinchilla(light gray)
337. ----------: ich genotype is not of the golden cost color in Labrador retriever?__bbEE
338. ---------- Building blocks of nucleic acida are ____Nucleotides
339. ---------- The numbers of genotypes can be calculeted on the basis of__Numbers of oteles
340. ----------70% genes in human are_____Mono-allelic
341. ----------The giant chromosomes found in and they may be as long as 300 µ and up to Diptera
342. ---------- How many number of bands can be recognize_____5000
343. ----------: two duplicated homologues separate during the an_____Meiosis I
344. ---------- Generally, karyotypes represent by arranging in the_____Descending straight line
345. ---------- Transmission of traits from parents to offspring_____Heredit
346. ---------- The nuclieone is composed of a core of histone proteins and_____Six 160
347. ----------: Whose exp.. result gives the the first solid evidence indicating___T h.morgen
348. ---------- Which of the following is not correct when gone is switched ON and transcription Acetylated histones
349. ----------: how many numbers of Allen can generale 10 genotypes?____Genes with 4 allelesS

……..

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FINAL TERM QUIZE

1. Genes will be switched on when histones Acetylated
2. vorinostat is histone deacetylase inhibitor
3. Exchange of DNA Segments between.chromosomes is known crossing over
4. Control of expression of gene by The expression of other Gene Epistasis
5. Expression of Phenotype of particular allele by number of individuals Expressivity
6. Centromere inverted in on of the following Chromesomal inverron Pericentric inversions
7. Human made Chromosomes is normally written as 46.XY
8. Chrome somal aberrations are classified into 2 types
9. Formation of Plenipotent cells from totipotent cells is called Cellular differentiation
10. Expression of "Dhene type of particular ailele by number Expressivity
11. Addition of a complete set chromosome is named as Euploidy
12. Two genes close together, frequency of crossing over would be Low
13. Large arm of chromosomes is ? arm
14. Colchicine to promoter Polyploidy
15. :Polygenic inheritance is controlled by Multiple genes
16. Fur color in Labrador is controlled by 2 genes
17. cancer, infertility are result of Translocation
18. : Which is following is not chromosomal aberration Translation
19. which is result of frame shift Deletion
20. Which is not true recessive trait Females are more affected
21. : Molecular biology, forensic , genes mapping are application of VNTR
22. About 25000 SNA in human are Non-Synapse
23. Determination of pattern inheritance Pedigree
24. Types of single nucleoide polymorphism that change the sequence of amino acid Non-coding
25. Expression of phenotype Incomplete inheritance
26. Small case for alphabets used for RNA level
27. Frequency of autosomal recessive 2-2.5

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28. Medelian disorder is Single gene
29. Addition mutation is addition of Nucliotide
30. Which is one of chromosomal duplication Retro - transposition
31. Turner syndrome is an example of Monosom
32. Mutation which type a mutant phenotype back into the wild type Reverse mutation
33. A genetic disease or disorder appears as a result of mutations in an individual' DNA
34. Hunter's disease is an example of_ X-linked recessive disorder
35. A condition in which some people with an appropriate genotype fail to express the phenotype is known as the incomplete penetrance
36. Which one of the following is not influenced by the silent mutation? Phenotype
37. Mutation that produce the change in the amino acid in protein product (histidine in for arginine Missense mutation
38. Base analog have the similar structure with standard bases of the DNA
39. Mutation which convert a DNA sequence into the stop codon. Non sense mutation
40. The term Ashkenazi refers to the specific people of following population.Jew population
41. ABO BLOOD group system is an example of multiple alleles
42. Gene maping give adea of gene location.
43. Which of following have single genotypes Blood group AB and O
44. Fruit clor of summer squish plant is an example Dominent epistatis
45. Pepper color is controlled 2 Gene
46. Position depended method of disease identification following Reverse genetics
47. Genetics can be studied by Two ways
48. Example of Mendilian genetics None
49. Condition in which an intermediate phenotype is developed from cross of parental generation. incomplete dominance
50. Condition in which one gene prevent the expression of other gene epitasis
51. chromosome is made up of DNA
52. structure for chromosome stability telomers
53. Broken ends of chromosomes are sticky
54. Centromere is unable to work due to mutation in CDE III
55. XO system of sex is present in grasshopper

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56. prokaryotic cell divides by binary fission
57. Chromosomes become fit in the central line metaphase
58. Haploid cells are produced by mitosis
59. Tetrads align at center metaphase
60. Genes are present on chromosomes first time said by greogr mendal
61. Cross for determination of genotype of an individual test cross
62. In ABO blood group system Alleles A and B show co-dominance
63. If two genes are more close then frequency of cross over will be high
64. In nature variation is Wide Spread.
65. Intermediate are produced in Incomplete Dominance.
66. Alleles at one gene hid/ prevent expression at second locus Epitasis.
67. Lung congestion occurs in Cystic fibrosis.
68. Cell spend 90% of its time in Interphase.
69. Chromosomes are stabilized by Telomere.
70. Nucleosomes folds and form Loops.
71. Which band is not found in Plants G-band.
72. Most common genetic markers Micro satellite SNPs
73. The possible arrangement of the these genes in which eye color is placed between wala the other two.
74. Cross b/w two for finding genome Test Cross
75. Mechanisms of genetics 2
76. Condition in which an organism is with 47 chromosomes with XXX triple X syndromes
77. Histone Deacetylase inhibitor is Vorionstat
78. When organism / or individual with allele for disease is unable to express the trait Incomplete Peiet .
79. What is not chromosomal aberration Translation.
80. Cancer, infertility leukemia are diseases of Translocation.
81. Chromosome 13 trisomy Patau's syndrome
82. Disease due to monosomy Turner's syndrome
83. Cochicine Promotes Polyploidy.

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84. Mutation in each cell 6
85. Mutation that pass onto next generation Green Line.
86. Organism with suppressor mutation is a Double mutant.
87. Point mutation effects single loci on DNA.
88. Effect DNA sequence but not protein: Silent mutation.
89. Mutations in with small letters are used for nucleotide RNA Level.
90. _____indicates affected residue.
91. SNP are found on coding & non coding region.
92. 25000 SNPS None synonymous.
93. Pedigree helps to determine mode of inheritance.
94. Frequency of autosomal recessive disorder is 2-2.5
95. In X-linked disorder all are true except Female are more affected than male.
96. When gene and environment both contributes Multifactorial.
97. About 25000 SNA in human are Non Synapse.
98. Expression of Phenotype Incomplete Inheritance
99. Mendilian disorder is single gone.
100. Addition mutation is addition of Nucleoside.
101. Types of single nucleoside polymorphism that change the sequence of amino acid Non-Coding.
102. Which mutation convert amino acid into stop codon non sense mutation.
103. Change in chromosome number is aneuploidy.
104. Type of trisomy which extra chromosomes arms derived from different chromosomes Tertiary Truism.
105. Which blood group do not carry any antigen O Blood Group.
106. Cytokines is occurs in Ten Phase.
107. The Physical location of an allele on a gene is Locus.
108. Sex determination chromosomal pattern in fishes is Z W system
109. The larger arm of the chromosome is named as P-arm
110. Medallion inheritance is Polygenic.
111. Genetic marker for a single locus is AFLP
112. Discontinous variation is Permanent.

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113. Which one has larger chromosome Plants.
114. No. of chromosome in Tobacco Plant 4
115. How many types of chromosomes banding 9
116. Histone Protein are basic bamino acid argentine & lysine
117. In Bacteria DNA is unwind by DNA Gyrase
118. DNA is left handed helix
119. Replication in lagging strand is discontinuous
120. Di-allelic gene generate genotype 3
121. DNA duplication caused by Ratio Transposition.
122. Which on is sexual disease Alzheimer's disease.
123. Phenotype depend on both.

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